JASON ZHI LIANG, Ph.D.

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Research Summary

Principal Research Scientist at Cognizant AI Lab, leading work on **automated AI research**, **open-ended learning**, and **self-improving agentic systems**, built on a decade of large-scale neuroevolution and evolutionary optimization for deep learning. I focus on automating the AI-research loop, from designing architectures and training recipes to knowledge synthesis, with minimal human intervention. Two convictions guide this work: a system's founding values compound over its lifetime, and open-endedness and creativity must continue at inference time, not just during training.

My work spans three layers of the AI stack: architectures (**CoDeepNEAT**, evolutionary neural architecture search; multitask extension, GECCO 2018), training recipes (**EPBT**, loss-function and hyperparameter evolution, GECCO 2021), and agents (**Caesar**, a deep-research agent, and **values-guided self-improving coding agents**). The same principle recurs at each level: propose candidates by search and select them with a score the candidate cannot influence, whether evolving architectures and training recipes below or refining an agent's own code and reasoning above. Named inventor on seven issued U.S. patents (first named inventor on six) in evolutionary AI and neural architecture search; **UT Austin CS Ph.D.** and **UC Berkeley EECS B.S.**

Technical Expertise

- **Evolutionary optimization & architecture search:** neuroevolution, neural architecture search, population-based training, multi-task learning, evolutionary loss/hyperparameter optimization.
- **Agentic & self-improving systems:** self-improving coding agents, open-ended exploration, deep-research agents, multi-agent orchestration, knowledge-graph reasoning, inference-time compute scaling, LLM-as-a-judge evaluation.
- **Distributed systems for search:** asynchronous evaluation and scheduling, cluster-scale parallel search.
- **Languages & frameworks:** Python, Claude Code, LLM APIs (Claude, Gemini, GPT), TensorFlow and Keras.

Professional Experience

Cognizant AI Lab | Principal Research Scientist

San Francisco, CA | Dec 2018 – Present | Lead Researcher driving the roadmap for the lab's automated AI research effort.

1. Autonomous Self-Improvement Systems

- **Values-Based Self-Improvement Framework:** Built a **self-improvement framework on Claude Code** in which coding agents improve their solver code over repeated sessions, carrying forward committed code under fixed guardrails against reward hacking, with a held-out

test set kept outside the agent's workspace. Showed that **the guiding values an agent starts with shape what it builds**. In a six-task study, a short values statement favoring exploration (*openness to change*) raised held-out scores over no-statement and placebo controls and led agents to try new algorithm families rather than tune parameters; in follow-up runs it also beat values the agents wrote for themselves and values distilled from AI papers (under submission to ICLR 2027).
- **Circle-Packing Records from Agent-Written Solvers:** In variable-radius circle packing, the framework's solvers matched the best-known value at **N=26** and exceeded the published **AlphaEvolve** and **ShinkaEvolve** packings under exact verification, broke David Cantrell's 2011/12 record at **N=27**, and now hold **459 of the 557** best-known values listed from N=27 to N=10,000 on Packomania (as of 1 November 2026).

2. Caesar: Creative Reasoning & Knowledge Synthesis

- **Automated Research via Inference-Time Self-Refinement:** Designed and implemented "Caesar," an autonomous deep-research agent that scales inference-time compute with a refinement loop: it critiques its own drafts, searches for what each draft is missing, and merges the findings into one answer. **Outperforms the strongest of three frontier deep-research agents by 13–23%** on a blinded creative-synthesis benchmark scored by an LLM judge.
- **Graph-Based Associative Reasoning:** Replaced fixed retrieval (RAG) with a loop that reads, reasons, and acts over a knowledge graph built during the search, surfacing cross-field connections that one-shot retrieval misses.
- **Open-Ended Exploration Policy:** Designed Caesar's exploration policy, which detects a stalled search and backtracks so the agent keeps finding new material over long runs. Cutting the exploration budget from **1,000 to 250 iterations** lowered long-form answer quality, suggesting that deep exploration of the graph drives the gain.

3. Multi-Agent Systems & Large-Scale Evolutionary Optimization

- **Multi-Agent Code Generation:** Developed a **multi-agent orchestration framework** in which specialized coding agents hand off work through defined stages, using evolutionary search to choose agent roles and team composition on the SciCode and HumanEval+ coding benchmarks.
- **Evolutionary Population-Based Training (EPBT):** Created a method that evolves the loss function and hyperparameters while the network trains. Beat the PBT baseline by **1.3 points** on CIFAR-10 ResNet-32 (92.79% vs 91.53%) and tested **520 candidate loss functions for the cost of 40 training runs, a 13× saving** (GECCO 2021; U.S. patent pending).
- **Code Repository Analysis:** Created "Code Archaeologist," an agentic system that reasons over repository histories (Git/LFS) to detect architectural patterns and technical debt.

Sentient Technologies | Research Scientist (prev. Research Intern)

San Francisco, CA | Dec 2015 – Nov 2018

- **Distributed Evolutionary Architecture Search:** Scaled **CoDeepNEAT** (the architecture-search method from my UT Austin Ph.D.) to GPU clusters, where candidate networks are trained and scored in parallel without waiting on one another. Productionized for image captioning with Sentient's product team.
- **Multi-Task Learning:** Set the state of the art at the time on the **20-task Omniglot multi-task benchmark (88% accuracy, up 21 points from the Soft Ordering baseline at 67%)** by evolving network structures that share components across tasks.

The University of Texas at Austin | Research Assistant / Ph.D. Candidate

Austin, TX | Sep 2013 – Dec 2018

- **Bilevel Meta-Evolution for Control:** Developed **MEA (Meta-Evolutionary Algorithm)**, which evolves the settings of the evolutionary algorithm itself, applied to continuous control (helicopter hovering, double pole balancing).

Selected Publications & Patents

Google Scholar: 2,464 citations, h-index 13 (1 November 2026)

Values Shape What Self-Improving Coding Agents Build (2026) ***Jason Liang***, *Dylan Waldner, Risto Miikkulainen*. Preprint. ICLR 2027 submission.

- Controlled study of how human value principles steer multi-session coding agents across six coding tasks, shaping both held-out scores and the algorithm families agents build.

Caesar: Deep Agentic Web Exploration for Creative Answer Synthesis (2026) ***Jason Liang***, *Elliot Meyerson, Risto Miikkulainen*. arXiv:2604.20855. Presented at the **Human-AI Co-Creativity Workshop, ICML 2026**.

- Autonomous research agent that scales inference-time compute through an adversarial refinement loop over a dynamic knowledge graph, outperforming the strongest of three frontier deep-research agents on a blinded, LLM-judged creative-synthesis benchmark.

Asynchronous Evolution of Deep Neural Network Architectures (2024) ***Jason Liang***, *Hormoz Shahrzad, Risto Miikkulainen*, Applied Soft Computing, Vol. 152.

- Techniques for asynchronous population-based optimization at scale, for efficient training on large distributed clusters.

Regularized Evolutionary Population-Based Training (2021) ***Jason Liang***, *Santiago Gonzalez, Hormoz Shahrzad, Risto Miikkulainen*, GECCO 2021.

- Method to dynamically optimize loss functions during training, improving convergence speed and generalization.

Evolutionary Neural AutoML for Deep Learning (2019) ***Jason Liang***, *Elliot Meyerson, Babak Hodjat et al.*, GECCO 2019.

- Evolutionary AutoML that jointly optimizes architecture, hyperparameters, and network size, not hyperparameters alone.

Evolutionary Architecture Search for Deep Multitask Networks (2018) ***Jason Liang***, *Elliot Meyerson, Risto Miikkulainen*, GECCO 2018.

- Extends CoDeepNEAT to multitask architecture search, automatically discovering deep multitask networks that outperform human-designed baselines.

Evolving Deep Neural Networks (2017/2019) *Risto Miikkulainen,* ***Jason Liang***, *Elliot Meyerson et al.*, Artificial Intelligence in the Age of Neural Networks and Brain Computing.

- Book chapter surveying neuroevolutionary methods for deep learning; introduces CoDeepNEAT, a coevolutionary neural architecture search algorithm (first named inventor, U.S.

11,003,994, 11,250,327 and 11,250,328); cited as prior work in Real et al., *Regularized Evolution* (AAAI 2019) and *AutoML-Zero* (ICML 2020).

Evolutionary Bilevel Optimization for Complex Control Tasks (2015) ***Jason Zhi Liang****, Risto Miikkulainen*, GECCO 2015.

- Bilevel evolutionary optimization of neuroevolution hyperparameters, applied to continuous-control tasks (helicopter hovering, double pole balancing).

Patents (Issued, U.S.):

- **U.S. Patent 12,406,188:** System and Method for Evolved Data Augmentation and Selection (2025).
- **U.S. Patent 12,282,845:** Multi-objective Coevolution of Deep Neural Network Architectures (2025).
- **U.S. Patent 11,507,844:** Asynchronous Evaluation Strategy for Evolution of Deep Neural Networks (2022).
- **U.S. Patent 11,250,328:** Cooperative Evolution of Deep Neural Network Structures (2022).
- **U.S. Patent 11,250,327:** Evolution of Deep Neural Network Structures (2022).
- **U.S. Patent 11,030,529:** Evolution of Architectures for Multitask Neural Networks (2021).
- **U.S. Patent 11,003,994:** Evolutionary Architectures for Evolution of Deep Neural Networks (2021).

Patents (Pending, U.S.):

- **App. 17/389,961:** Regularized Evolutionary Population-Based Training (2023).
- **App. 17/064,706:** Sharing Meta-Learning Methods Among Multiple Private Data Sets (2022).

Awards & Professional Activities

- **BEACON Research Grant** (2015), BEACON Center for the Study of Evolution in Action (an NSF Science & Technology Center).
- **Silver Reviewer**, International Conference on Machine Learning (ICML)
- **Reviewer**, Genetic and Evolutionary Computation Conference (GECCO)
- **Talk:** *The Next Frontier in AI: Multi-Agent Systems*. Hodjat, Singh, Liang, Qiu. NVIDIA GTC 2025.

Education

Ph.D. in Computer Science · *Neuroevolution & Deep Learning*
The University of Texas at Austin · Sep 2013 – Dec 2018 · Advisor: Risto Miikkulainen
Dissertation: *Evolutionary Neural Architecture Search for Deep Learning*

B.S. in Electrical Engineering and Computer Science
University of California, Berkeley · Sep 2009 – May 2013